Robotics Program

Facilitator Guide

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Introduction

This document is showing the list of experiments and procedures which will be conducted by students in this course

As Facilitator you should be in advance in the following topics:

Course No1: Introduction to Electricity and Electronics

Depending on students ages you will conduct the following topics in this course

Week 1: Overview of Electricity and Circuits

In this week you will explain the following topics with practical example:

a) Electricity in general
b) Types of electricity (DC and AC)
c) Electricity effects
d) Electricity polarity
e) Safety in electricity
f) Batteries and power supplies

Week 2: DC Circuits I (Simulation Mode/Practical Mode)

In this week you will explain the following topics with practical example:

a) Battery and power supplies
b) Conductor & nonconductor Materials
c) Simple circuit experiment
d) Experiment of Conductor & nonconductor in simple circuit (cut a wire and place different material in a gap do demonstrate Conductor & nonconductor materials)
e) The result you should show that the voltage of the power supply is equal to the voltage across the lamp

Tools :
1) https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_all.html
2) 12v Power supply, wires, switch, 12v lamp

Week 3: DC Circuits II (Simulation Mode/ Practical Mode)

In this week you will explain the following topics with practical example:

a) A serial Circuit
b) Calculation in simulation Mode of voltage across each lamp in the circuit using Multimeter
c) The student should know that the summation of **all voltages across lords is equal to the voltage source and the current is constant in all the circuit**

Week 4: DC Circuits III (Simulation Mode/ Practical Mode)

In this week you will explain the following topics with practical example:

Using previous knowledge students will make:

a) A parallel circuit
b) Calculation in simulation Mode of current in branches of the circuit

   NB: if they are able to calculate show them Ohm's Low and Kickoff Low else let them do practical only
c) Student should do a prototype of a **Simple Home Installation which has 3,4,5 Rooms** depending on available materials and number of students

Course Practical's List

If you have more this allow your students to do following practical's:

a) Simple circuit with lamp and Motor
b) Parallel Circuit with lamps and Motors
c) Serial Circuit with Motors
d) Contraction and installation of a prototype of a **Simple Home Installation which has 3,4,5 Rooms**
e) Allow them to design and do there own electrical installation